Computer-Aided Design of Digital Systems (CAD)

ECE536(CAD), Lecture 2:

Logic Design Overview

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Modern Digital Systems

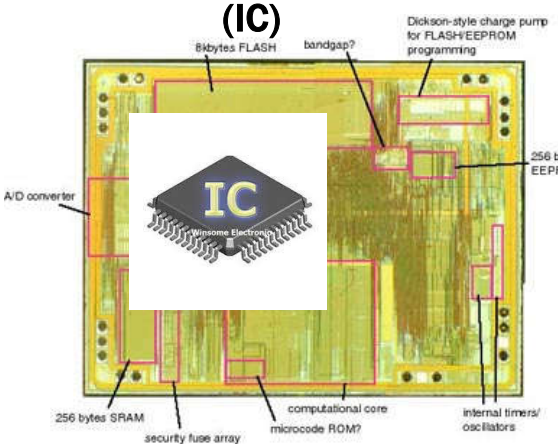
Personal computer (PC)



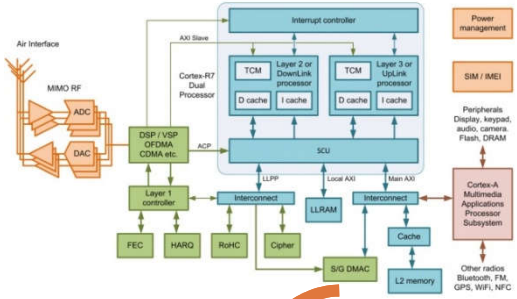
Printed circuit board (PCB)



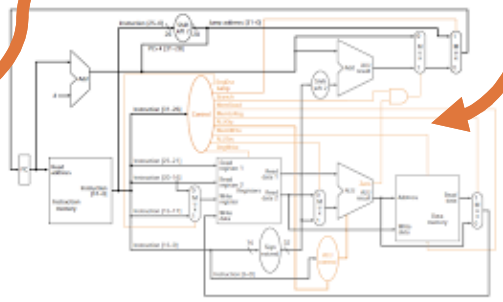
Integrated circuit (IC)



Modules

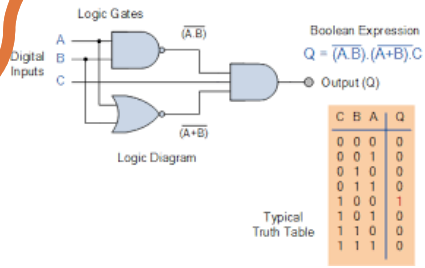


Cells



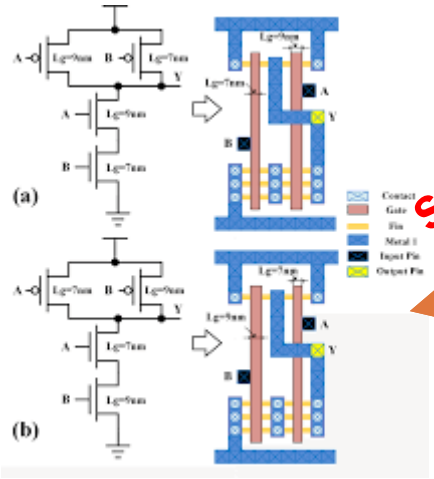
Synthesis

Gates



Synthesis

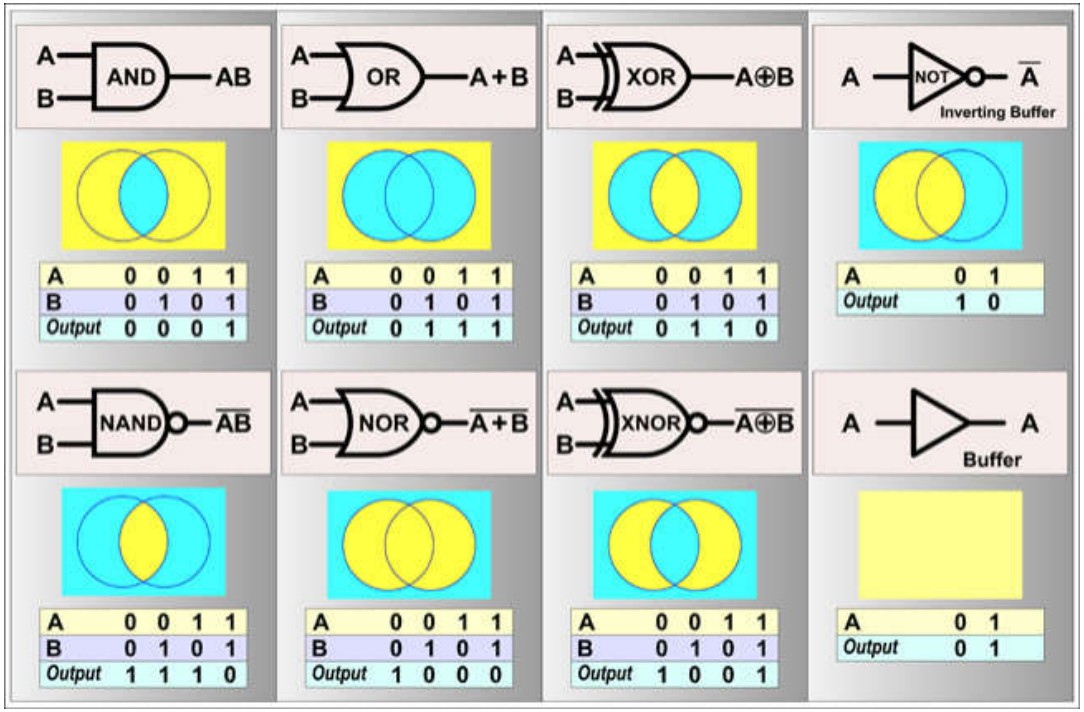
MOSFET transistors



Synthesis

Logic Gates

- **Gate**
  - The manipulation of binary information is done by logic circuit called "gate".
- ***Digital Logic Gates***
  - AND, OR, INVERTER, BUFFER, NAND, NOR, XOR, XNOR



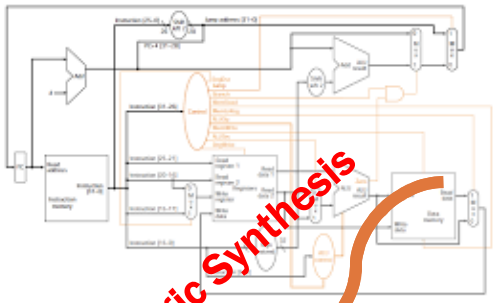
Digital Design Flow

```
module fa(
input a, b ,ci,
output s, co);
wire [1:0] t ;
assign t = a+b+ci;
assign co = t[1];
assign s = t[0];
endmodule
```

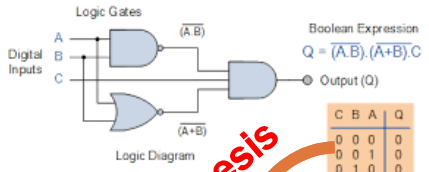
Logic Synthesis

```
module fa ( a, b, ci, s, co );
  input a, b, ci;
  output s, co;
  wire   n1, n2, n3, n4, n6, n7;

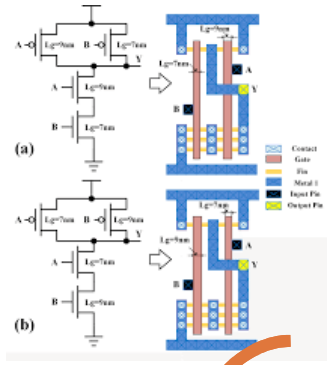
  MUX2X1 U3 ( .B(n2), .A(ci), .S(n3), .Y(s) );
  MUX2X1 U4 ( .B(n1), .A(b), .S(a), .Y(n3) );
  NAND2X1 U5 ( .A(n4), .B(n7), .Y(co) );
  OAI21X1 U7 ( .A(b), .B(ci), .C(a), .Y(n4) );
  INVX1 U2 ( .A(ci), .Y(n2) );
  INVX1 U1 ( .A(b), .Y(n1) );
  AND2X1 U8 ( .A(b), .B(ci), .Y(n6) );
  INVX1 U9 ( .A(n6), .Y(n7) );
endmodule
```



Logic Synthesis



Physical Synthesis

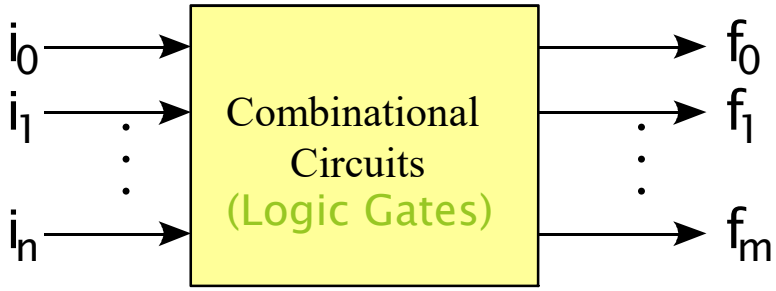


Fabrication



Combinational Circuits

connected arrangement of *logic gates* with a set of inputs and outputs



- **Analysis**
  - Logic circuits diagram → Boolean function or Truth table
- **Design** (Analysis)
  - 1. The Problem is stated
  - 2. I/O variables are assigned
  - 3. Truth table(I/O relation)
  - 4. Simplified Boolean Function
  - 5. Logic circuit diagram

Logic Simplification

- Karnaugh Map(K-Map)
  - Map method for simplifying Boolean expressions
- Minterm / Maxterm
  - Minterm : n variables *product* (x=1, x'=0)
  - Maxterm : n variables *sum* (x=0, x'=1)
- 2 variables example

x	y	Minterm	Maxterm
0	0	x'y' m_0	x + y M_0
0	1	x'y m_1	x + y' M_1
1	0	x y' m_2	x'+ y M_2
1	1	x y m_3	x'+ y' M_3

m0 + m1 + m2 + m3

M0 • M1 • M2 • M3

K-Maps

- 2 variables

A \ B	0	1
2	3	

- 3 variables

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

- 4 variables

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

- 5 variables

A \ BCDE	000	001	011	010	110	111	101	100
00	0	1	3	2	6	7	5	4
01	8	9	11	10	14	15	13	12
11	24	25	27	26	30	31	29	28
10	16	17	19	18	22	23	21	20

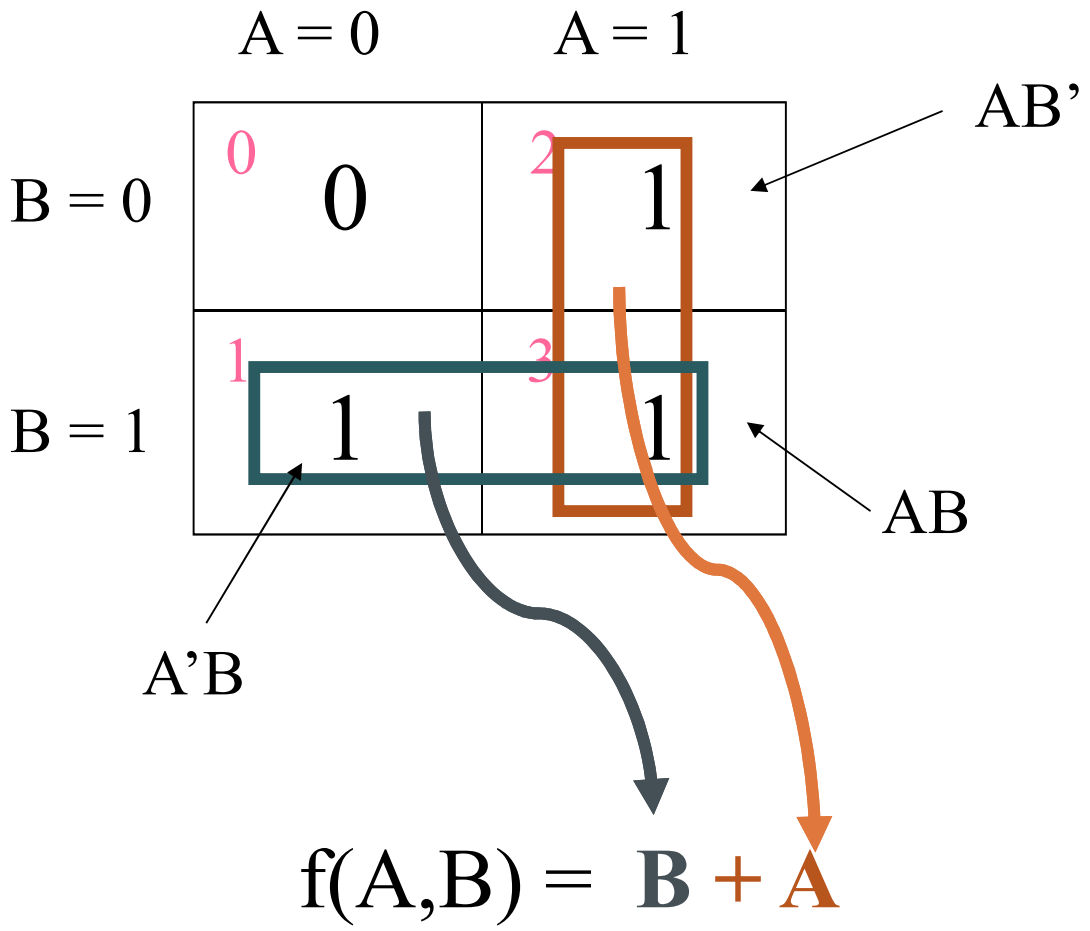
Boolean Theorems: Summary

	Theorem		Dual	Name
T1	$B \bullet 1 = B$	T1′	$B + 0 = B$	Identity
T2	$B \bullet 0 = 0$	T2′	$B + 1 = 1$	Null Element
T3	$B \bullet B = B$	T3′	$B + B = B$	Idempotency
T4		$\overline{\overline{B}} = B$		Involution
T5	$B \bullet \overline{B} = 0$	T5′	$B + \overline{B} = 1$	Complements

	Theorem		Dual	Name
T6	$B \bullet C = C \bullet B$	T6′	$B + C = C + B$	Commutativity
T7	$(B \bullet C) \bullet D = B \bullet (C \bullet D)$	T7′	$(B + C) + D = B + (C + D)$	Associativity
T8	$(B \bullet C) + B \bullet D = B \bullet (C + D)$	T8′	$(B + C) \bullet (B + D) = B + (C \bullet D)$	Distributivity
T9	$B \bullet (B + C) = B$	T9′	$B + (B \bullet C) = B$	Covering
T10	$(B \bullet C) + (B \bullet \overline{C}) = B$	T10′	$(B + C) \bullet (B + \overline{C}) = B$	Combining
T11	$(B \bullet C) + (\overline{B} \bullet D) + (C \bullet D)$ $= B \bullet C + \overline{B} \bullet D$	T11′	$(B + C) \bullet (\overline{B} + D) \bullet (C + D)$ $= (B + C) \bullet (\overline{B} + D)$	Consensus
T12	$\overline{B_0 \bullet B_1 \bullet B_2 \ldots}$ $= (\overline{B_0} + \overline{B_1} + \overline{B_2} \ldots)$	T12′	$\overline{B_0 + B_1 + B_2 \ldots}$ $= (\overline{B_0} \bullet \overline{B_1} \bullet \overline{B_2})$	De Morgan's Theorem

On the K-map

f(A,B) = A'B +AB + AB' + AB
= B (A+A') + A (B'+B)
= B + A



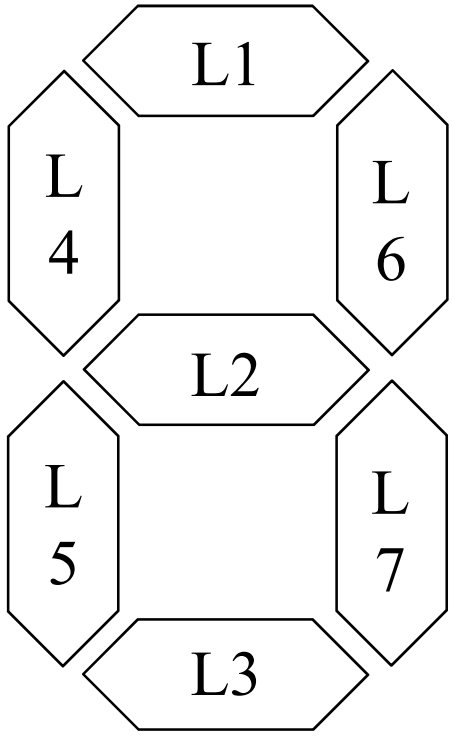
On the K-map

A \ B	0	1
0	0	2
1	1	3

C \ AB	00	01	11	10
0	0	2	6	4
1	1	3	7	5

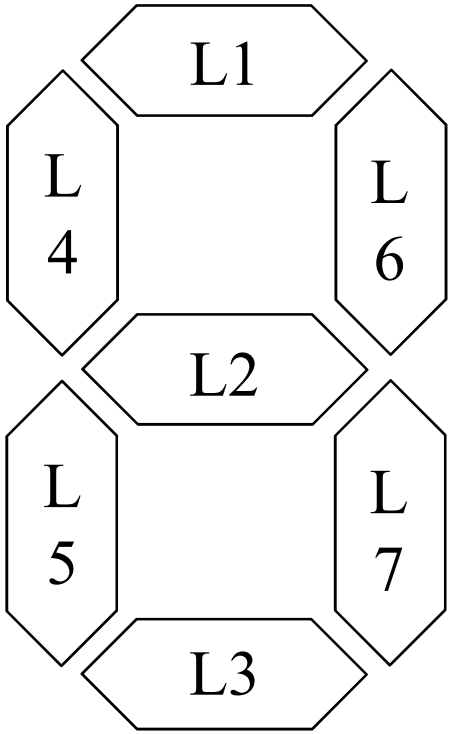
CD \ AB	00	01	11	10
00	0	4	12	8
01	1	5	13	9
11	3	7	15	11
10	2	6	14	10

Case Study of a Simple Logic Design: Seven Segment Display

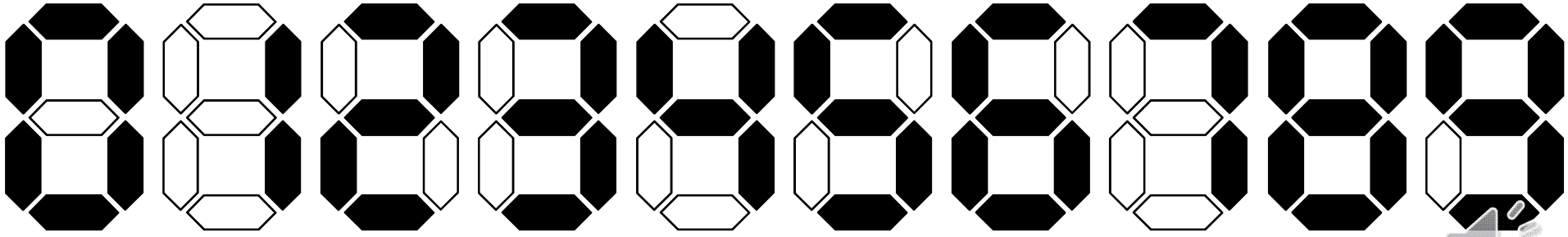


B3	B2	B1	B0	Val
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9

Case Study (cont.)



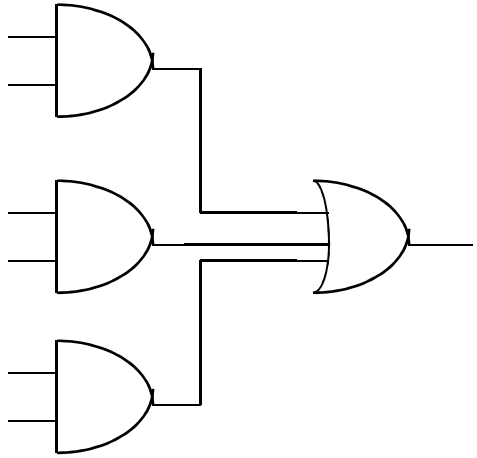
B3	B2	B1	B0	Val	L1	L2	L3	L4	L5	L6	L7
0	0	0	0	0	1	0	1	1	1	1	1
0	0	0	1	1	0	0	0	0	0	1	1
0	0	1	0	2	1	1	1	0	1	1	0
0	0	1	1	3	1	1	1	0	0	1	1
0	1	0	0	4	0	1	0	1	0	1	1
0	1	0	1	5	1	1	1	1	0	0	1
0	1	1	0	6	1	1	1	1	1	0	1
0	1	1	1	7	1	0	0	0	0	1	1
1	0	0	0	8	1	1	1	1	1	1	1
1	0	0	1	9	1	1	1	1	0	1	1



Case Study (cont.)

- **Implement L4:**

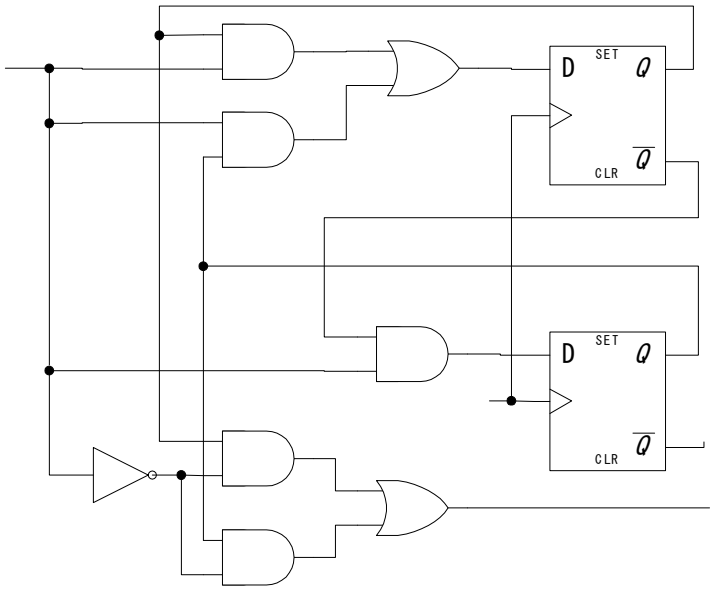
B3	B2	B1	B0	L4
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1

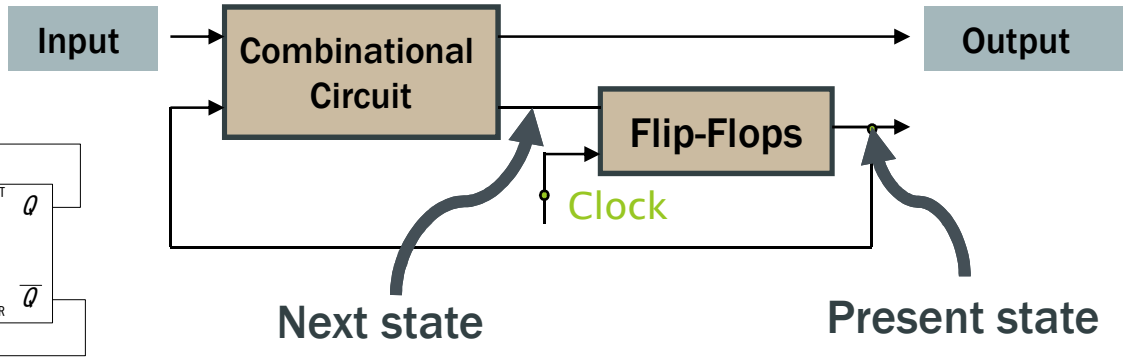


Some gate level implementation of the Boolean function for L4

Sequential circuits

- A sequential circuit is an interconnection of FFs and Gates

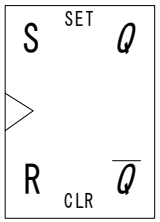




Flip-Flops

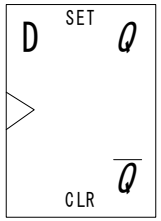
- The *storage elements* employed in clocked *sequential circuit*
- A binary cell capable of storing one bit of information

SR *(Set/Reset)* F/F



S	R	Q(t+1)	
0	0	Q(t)	no change
0	1	0	clear to 0
1	0	1	set to 1
1	1	?	Indeterminate

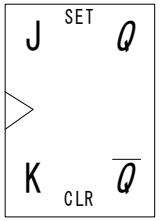
D *(Data)* F/F



D	Q(t+1)	
0	0	clear to 0
1	1	set to 1

- Q(t+1)=D

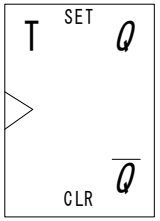
JK *(Jack/King)* F/F



J	K	Q(t+1)	
0	0	Q(t)	no change
0	1	0	clear to 0
1	0	1	set to 1
1	1	Q(t)'	Complement

- JK F/F is a refinement of the SR F/F
- The indeterminate condition of the SR type is defined in complement

T *(Toggle)* F/F



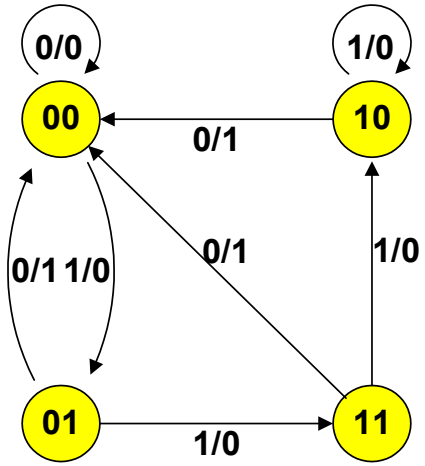
T	Q(t+1)	
0	Q(t)	no change
1	Q'(t)	Complement

- T=1(J=K=1), T=0(J=K=0) JK F/F
- Q(t+1)= Q(t) ⊕ T

Sequential circuit Design

State Diagram

- Graphical representation of state table
  - Circle(state), Line(transition), I/O(input/output)



State Table

- Present state, input, next state, output

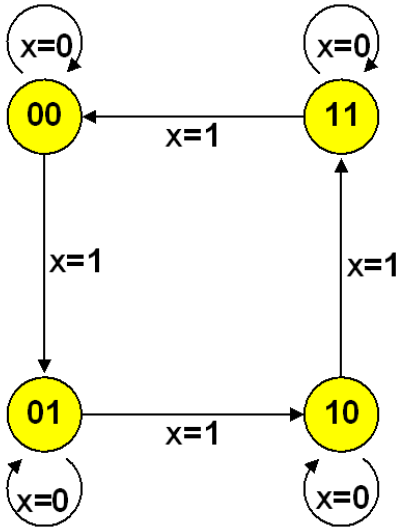
Present State		Input			Input Equ.		Next State		Output
A	B	x	Ax	Bx	D_A	D_B	A	B	y
0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	1	0
0	1	0	0	0	0	0	0	0	1
0	1	1	0	1	1	1	1	1	0
1	0	0	0	0	0	0	0	0	1
1	0	1	1	0	1	0	1	0	0
1	1	0	0	0	0	0	0	0	1
1	1	1	1	1	1	0	1	0	0

Excitation Table

JK F/F			
Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Binary Counter

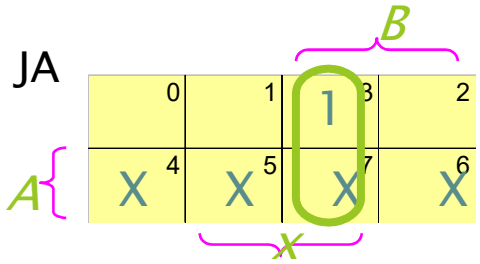
x=1: 00, 01, 10, 11, 00, ...
x=0: no change



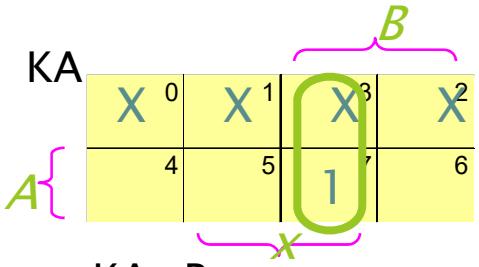
Present State		Input	Next State		F/F Input			
A	B	x	A	B	J_A	K_A	J_B	K_B
0	0	0	0	0	0	x	0	x
0	0	1	0	1	0	x	1	x
0	1	0	0	1	0	x	x	0
0	1	1	1	0	1	x	x	1
1	0	0	1	0	x	0	0	x
1	0	1	1	1	x	0	1	x
1	1	0	1	1	x	0	x	0
1	1	1	0	0	x	1	x	1

JK F/F

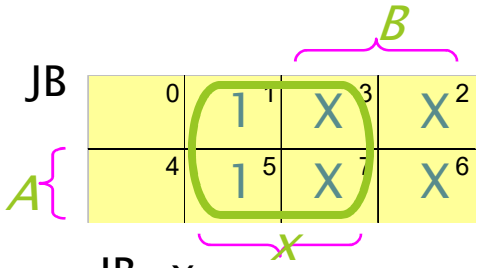
Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



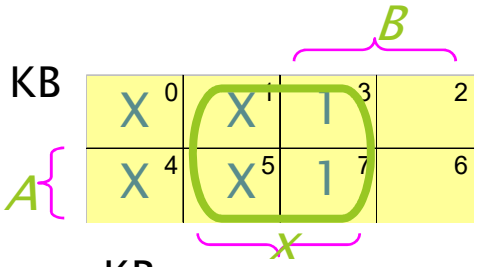
JA=Bx



KA=Bx



JB=x

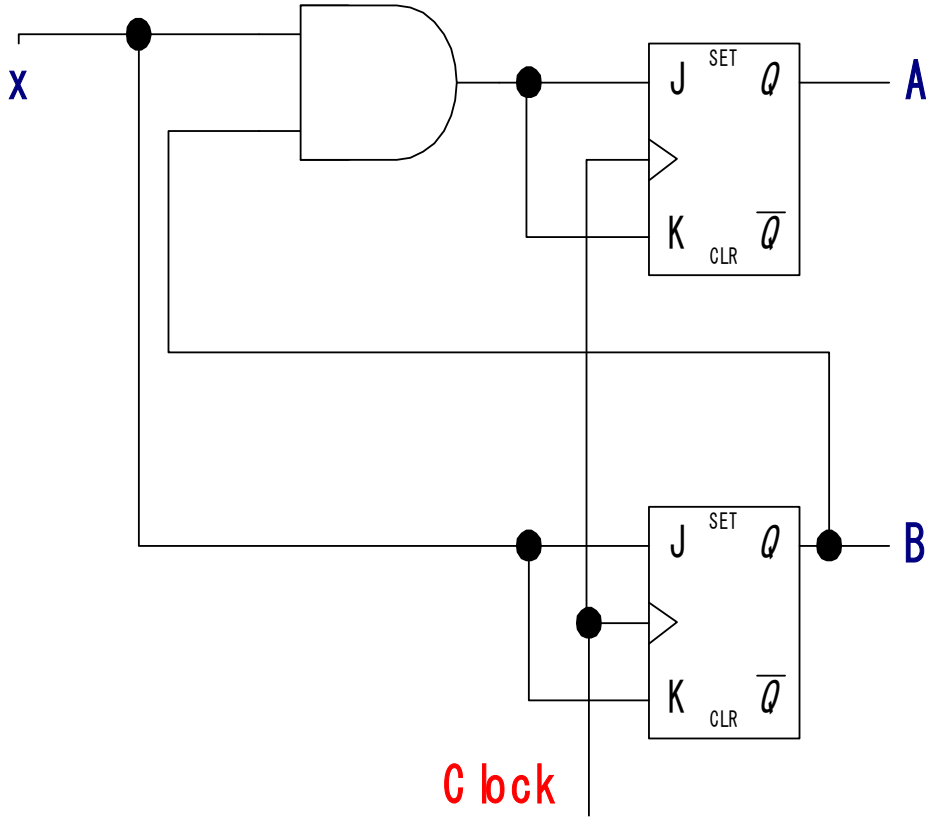


KB=x

Binary Counter With JK FF

JA=Bx KA=Bx

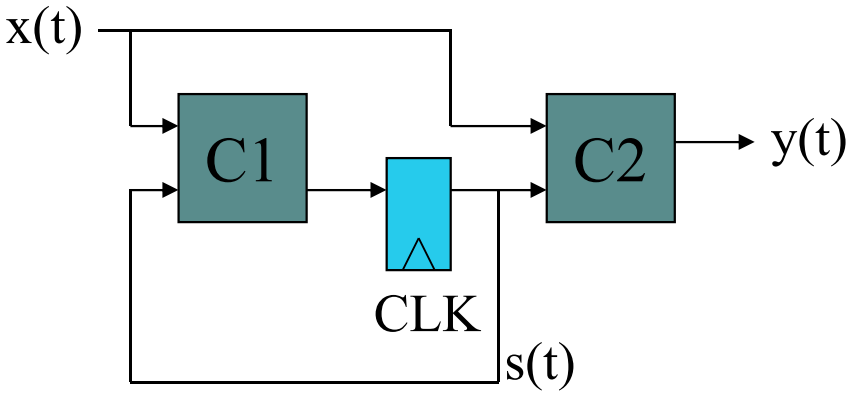
JB=x KB=x



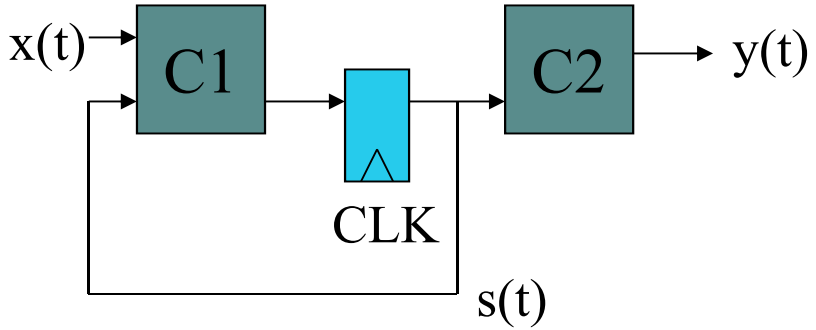
Canonical Form: Mealy and Moore Machines

$s_i(t+1) = g_i(X(t), S(t))$

Mealy Machine: $y_i(t) = f_i(X(t), S(t))$
Moore Machine: $y_i(t) = f_i(S(t))$



Mealy Machine



Moore Machine

?